MATH 10A — STRUCTURE AND CONCEPTS IN MATH I

§1.3 The Roman Numeration System

Chapter 1 — Numeration Systems

Today's Goal

After completing this section, you should be able to:

- Read and write numbers using the Roman numeration system.

LT.2 Numeration Systems

I can convert numbers between various numeration systems including Roman, Egyptian, Mayan, Babylonian, and Hindu-Arabic.

What Year Is On the Cornerstone?

A building's cornerstone reads:

M M X X V I

What year is that?

M is $1{,}000$, X is 10, V is 5, I is 1. Add them up:

$$1{,}000 + 1{,}000 + 10 + 10 + 5 + 1 = 2{,}026.$$

You added. That is most of the Roman system — but not all of it.

Seven Symbols, Seven Values

Roman Numeral	Number
I	1
V	5
X	10
L	50
C	100
D	500
M	1,000

Mostly, You Just Add

For the most part, the Roman system is an **additive numeration system** (Definition 1.2.2): the value of a set of glyphs is the sum of the values of each glyph.

- XXVII is $10 + 10 + 5 + 1 + 1 = 27$.
- DCLXIII is $500 + 100 + 50 + 10 + 1 + 1 + 1 = 663$.

The symbols are written largest first, then down. Check the second one yourself.

How Would You Write Four?

You have only I and V.
Write the number 4.

IIII does add to 4 — but the Romans did not write it that way. They avoided repeating a symbol more than three times.

Instead they wrote **IV**, and read it as $5-1$.

A small symbol placed *before* a big one means subtract.

The Subtractive Property

Definition 1.3.3

The **subtractive property** of the Roman numeration system states that when a smaller numeral is placed directly before a larger one, the smaller numeral is subtracted from the larger one rather than added. The only allowed pairings are $\mathrm{IV} = 4, \mathrm{IX} = 9, \mathrm{XL} = 40, \mathrm{XC} = 90, \mathrm{CD} = 400,$ and $\mathrm{CM} = 900.$

The Six Subtractive Pairs

Applying the property to the numerals in Table 1.3.1 gives exactly six pairs.

Roman Numeral	Number
IV	$5-1=4$
IX	$10-1=9$
XL	$50-10=40$
XC	$100-10=90$
CD	$500-100=400$
CM	$1{,}000-100=900$

Only Those Six

Those are the **only** combinations that use the subtractive property.

- IC is **not** $100 - 1 = 99$. It is not a Roman numeral at all.
- Ninety-nine is XCIX — that is 90 and 9, two legal pairs side by side.

How to spot a subtractive pair while reading: the values **go up** as you read left to right.

The Multiplicative Property

Definition 1.3.4

The **multiplicative property** of the Roman numeration system states that a horizontal bar (called a **vinculum**) placed over a Roman numeral multiplies that numeral's value by $1{,}000$. For example, $\overline{\mathrm{VII}} = 7{,}000$ and $\overline{\mathrm{LX}} = 60{,}000$.

So instead of writing MMMMMMM, we can write $\overline{\mathrm{VII}}$. Either is acceptable.

Reading a Roman Numeral

Procedure 1.3.5. Converting a Roman Numeral to the Corresponding Hindu-Arabic Numeral.

To convert a Roman numeral to the corresponding Hindu-Arabic numeral, do the following.

1. Examine the Roman numeral for any instances where the multiplicative property is needed and determine their value(s).
2. Examine the Roman numeral for any instances where the subtractive property is needed and determine their value(s).

Reading a Roman Numeral, continued

> **Procedure 1.3.5, continued.**
>
> 3. Determine the values for all other Roman numerals.
> 4. Add all the values from previous steps.

Steps 1 and 2 pull out the parts that are not plain addition. Steps 3 and 4 finish the job.

Reading a Numeral That Only Adds

What number is DCCXXXV?

No bar, and the values never go up. Just group and add.

Roman Numeral	Number
D	500
CC	$2 \cdot 100 = 200$
XXX	$3 \cdot 10 = 30$
V	5

Adding these gives the Hindu-Arabic numeral 735.

Reading Across Two Subtractive Pairs

What number is MMCCXLIX?

Still no bar. The value goes up twice: X to L, and I to X. Two subtractive pairs.

Roman Numeral	Number
MM	$2 \cdot 1{,}000 = 2{,}000$
CC	$2 \cdot 100 = 200$
XL	$50 - 10 = 40$
IX	$10 - 1 = 9$

Adding these gives the Hindu-Arabic numeral $2{,}249$.

Reading a Numeral With a Bar

What number is $\overline{\mathrm{XI}}\mathrm{CMXCIV}$?

The bar covers XI. After it, three values go up, so three pairs subtract.

Roman Numeral	Number
$\overline{\mathrm{XI}}$	$11 \cdot 1{,}000 = 11{,}000$
CM	$1{,}000 - 100 = 900$
XC	$100 - 10 = 90$
IV	$5 - 1 = 4$

Adding these gives the Hindu-Arabic numeral $11{,}994$.

Now You Try: Reading

You Try 1.3.1. Break MMMCXLIX into its parts, give the value of each part, then add.

Roman Numeral	Number
MMM	$3 \cdot 1{,}000 = 3{,}000$
C	100
XL	$50 - 10 = 40$
IX	$10 - 1 = 9$

Total. $3{,}000 + 100 + 40 + 9 = 3{,}149.$

Writing a Roman Numeral

Procedure 1.3.7. Converting a Hindu-Arabic Numeral to the Corresponding Roman Numeral.

To convert a Hindu-Arabic numeral to the corresponding Roman numeral, do the following.

1. Break the number into thousands, hundreds, tens, and ones.
2. Convert each component separately, making use of the subtractive property as needed.

Writing a Roman Numeral, continued

Procedure 1.3.7, continued.

3. If the number is larger than $3{,}999$, consider using the multiplicative property to write the number of thousands.
4. Combine the results from the conversion of each component.

Step 1 is expanded form. You have been doing it since §1.1.

Writing by Place Value

Write 857 as a Roman numeral.

Start with expanded form:

$$857 = 800 + 50 + 7.$$

- $800 = 500 + 3 \cdot 100$, using D and C: DCCC.
- 50 is L.
- $7 = 5 + 2$, which is VII.

Combining these gives $857 = \mathrm{DCCCLVII}$.

Writing When Subtraction Is Needed

Write $2{,}594$ as a Roman numeral.

In expanded form:

$$2{,}594 = 2{,}000 + 500 + 90 + 4.$$

- $2{,}000 = 2 \cdot 1{,}000$ is MM, and 500 is D.
- 90 uses the subtractive property: XC.
- 4 uses the subtractive property: IV.

Combining these gives $2{,}594 = \mathrm{MMDXCIV}$.

Writing a Number in the Millions

Write $2{,}780{,}946$ as a Roman numeral.

This number is in the millions, so the multiplicative property will keep the numeral short. Write it in "partially" expanded form:

$$2{,}780{,}946 = 2{,}780{,}000 + 900 + 40 + 6.$$

- 900 uses the subtractive property: CM.
- 40 uses the subtractive property: XL.
- $6 = 5 + 1$, so it is VI.

Putting the Bar to Work

That leaves $2{,}780{,}000$, which is $2{,}780 \cdot 1{,}000$. So write $2{,}780$ first.

$$2{,}780 = \underbrace{2{,}000}_{\mathrm{MM}} + \underbrace{700}_{\mathrm{DCC}} + \underbrace{80}_{\mathrm{LXXX}}.$$

So $2{,}780 = \mathrm{MMDCCLXXX}$. A vinculum over it multiplies by $1{,}000$:

$$\overline{\mathrm{MMDCCLXXX}} = 2{,}780{,}000.$$

$$2{,}780{,}946 = \overline{\mathrm{MMDCCLXXX}}\mathrm{CMXLVI}.$$

Now You Try: Writing

You Try 1.3.2. Write $2{,}049$ as a Roman numeral, one place at a time.

Place	Value	Roman Numeral
Thousands	$2{,}000$	MM
Hundreds	0	nothing to write
Tens	40	XL
Ones	9	IX

Combined. $2{,}049 = \mathrm{MMXLIX}$.

Three Rules, One System

- **Add.** When the values decrease as you read left to right, add them. XXVII is 27.
- **Subtract.** A smaller symbol directly before a larger one — six pairs only. XL is 40.
- **Multiply.** A bar over a numeral multiplies it by $1{,}000$. $\overline{\mathrm{XI}}$ is $11{,}000$.

Notice what a Roman symbol does *not* do: its value never depends on where it sits. That is what separates an additive system from the place value system of §1.1.